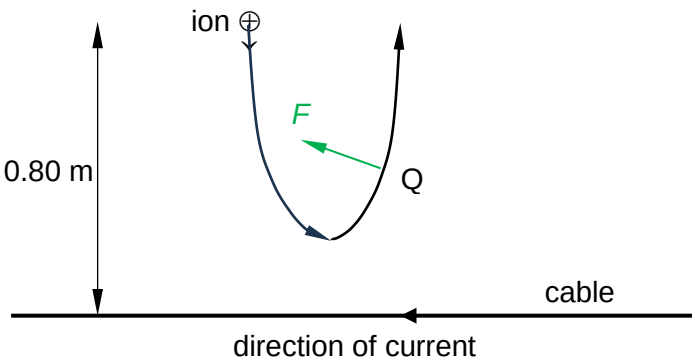
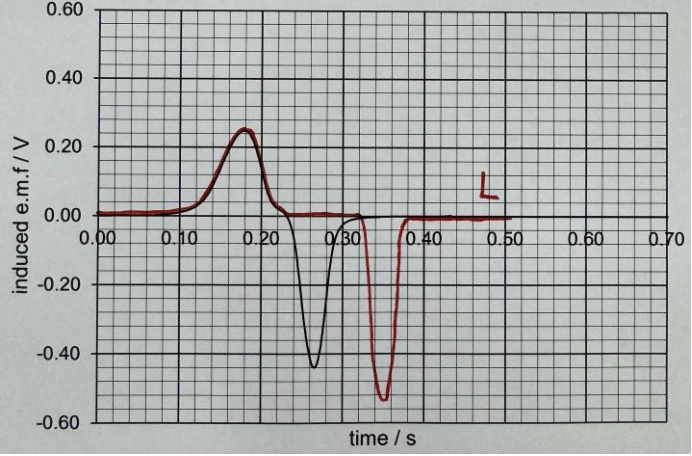
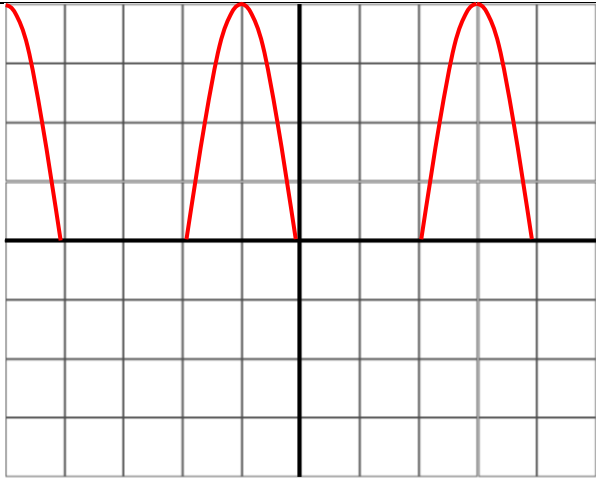


6	
(a)	Force per unit length acting on a straight, current-carrying conductor,
	carrying unit current and placed at right angles to an external magnetic field.

(b)(i)	$B = \left(\frac{4\pi \times 10^{-7} \times 80}{2\pi \times 0.80} \right)$
	$F = Bqv \sin 90^\circ$ $= \left(\frac{4\pi \times 10^{-7} \times 80}{2\pi \times 0.80} \right) (1.6 \times 10^{-19})(1.0 \times 10^3)$
	$= 3.2 \times 10^{-21} \text{ N}$
(b)(ii)	 Force drawn perpendicularly to the tangent of the path at Q.
(b)(iii)	At P, the ion will experience a magnetic force to the right by FLHR. This will make the ion move in a circle.
	As ion moves nearer the cable, magnetic flux density due to the cable increases ($B \propto 1/r$), increasing the magnetic force ($F = Bqv$).
	Thus, the radius of circular path becomes smaller. Eventually there will be a point where the ion is moving parallel to the cable. The magnetic force will then cause the ion to turn back, with the radius of the circular path becoming larger as the ion moves further away from the cable.
(c)(i)1.	The magnetic field of the magnet is not uniform and the magnet's velocity is changing.
	The rate of change of magnetic flux linkage will not be constant.
(c)(i)2.	The magnetic flux linkage increases when the magnet enters the coil and decreases when the magnet leaves the coil.
(c)(i)3.	The magnet's velocity increases as it falls through the coil.
	The magnitude of the change in the magnetic flux linkage (area under the graph) is the same.
(c)(ii)	

	The positive e.m.f. region will be the same (as the magnet is released from the same height and would have the same velocity entering the coil (same rate of change of magnetic flux linkage)). There will be a region where e.m.f. is zero. (This is when the magnet is moving fully inside the coil as there is no rate of change magnetic flux linkage).
	The negative e.m.f. region will have a larger value of maximum e.m.f. with a shorter duration (as the magnet would have a higher velocity leaving the coil due to greater distance moved (greater rate of change of magnetic flux linkage)). (The positive area should be roughly equal to negative area.)
(d)(i)	For a half-rectified wave, $I_{rms} = \frac{I_0}{2}$
	$V_0 = I_0 R$ $= (2 \times 5.0 \times 10^{-3})(2.0 \times 10^3)$ $= 20 \text{ V}$
(d)(ii)	
	Timebase: $f = 50 \text{ Hz}$, $T = 20 \text{ ms}$ (4 squares) Voltage gain: $V_0 = 20 \text{ V}$ (4 squares)
	Correct shape: half-rectified wave of at least 2 cycles
7	